

TravelTide Customer Segmentation - Executive Summary

Project Lead: Alberto Diaz Durana | **Date:** October 30, 2025 | **For:** Elena, Head of Marketing

TravelTide seeks to launch a personalized rewards program by assigning 5,765 active customers to one of five perks. Using hierarchical clustering across 65 behavioral features, we identified 3 distinct segments and assigned each user to their optimal perk via propensity scoring.

Three Customer Segments Identified

Premium Paula (79.7% | 4,596 users | \$4,985 avg CLV) - High-value hotel travelers driving 98% of total CLV (\$22.9M). Prefers premium hotel perks.

Dining David (5.0% | 287 users | \$768 avg CLV) - Mid-value experiential travelers who prioritize dining. Strong hotel meal preference.

Flexible Fiona (15.3% | 882 users | \$319 avg CLV) - Budget-conscious occasional travelers valuing flexibility over premium features.

Key Finding: Balanced Perk Distribution - Despite 79.7% cluster concentration, individual propensities create balanced perk allocation:

Free Bag:	24.3% (1,402 users)
Hotel Meal:	23.4% (1,349 users)
Hotel Night:	22.2% (1,277 users)
Exclusive Discount:	16.4% (944 users)
No Cancel Fee:	13.8% (793 users)

Critical Insight: All 5 perks viable - individual preferences within segments more diverse than cluster averages suggested.

Assignment Confidence: 97.2% of assignments enable confident implementation with minimal risk.

- **HIGH:** 63.5% (3,658 users) with strong preferences
- **MEDIUM:** 33.8% (1,947 users) with reliable preferences
- **LOW:** 2.8% (160 users) requiring A/B testing

Recommendations

- Deploy all 5 perks - Balanced distribution validates complete rewards suite
- Prioritize Premium Paula - Allocate 70-75% of marketing budget to retain high-value segment (\$22.9M CLV)
- Phased rollout - Launch HIGH confidence users (63.5%) first, expand to MEDIUM (33.8%), A/B test LOW (2.8%)

Expected Business Impact

- Revenue Protection: 15-20% churn reduction in Premium Paula = \$3.4M-\$4.6M protected annual revenue
- Marketing Efficiency: Personalized perks increase booking frequency 15%+ and improve satisfaction scores
- Competitive Advantage: Data-driven individual assignment differentiates from generic competitor loyalty programs

Bottom Line: Segmentation complete with 97.2% high-confidence assignments. All 5 perks validated. Ready for immediate launch with minimal risk and strong ROI potential.

Deliverables:

- user_perk_assignments.csv - Individual perk assignments with confidence scores (5,765 × 8)
- cluster_profiles_k3.csv - Segment characteristics and CLV metrics (3 × 10)
- cluster_assignments_k3.csv - User-to-cluster mapping (5,765 × 2)
- perk_distribution.csv - Summary of perk allocation across all users (5 × 3)

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